

TAXI: Evaluating Categorical Knowledge Editing for Language Models

Derek Powell[△] Walter Gerych[◇] Thomas Hartvigsen[♡]
[△]Arizona State University [◇]MIT [♡]University of Virginia
 dmpowell@asu.edu, wgerych@mit.edu, hartvigsen@virginia.edu

Abstract

Humans rarely learn one fact in isolation. Instead, learning a new fact induces knowledge of other facts about the world. For example, in learning a *korat* is a type of cat, you also infer it is a mammal and has claws, ensuring your model of the world is *consistent*. Knowledge editing aims to inject new facts into language models to improve their factuality, but current benchmarks fail to evaluate consistency, which is critical to ensure efficient, accurate, and generalizable edits. We manually create TAXI, a new benchmark dataset specifically created to evaluate consistency in categorical knowledge edits. TAXI contains 11,120 multiple-choice queries for 976 edits spanning 41 categories (e.g., Dogs), 164 subjects (e.g., Labrador), and 183 properties (e.g., is a mammal). We then use TAXI to evaluate popular editors' categorical consistency, measuring how often editing a subject's category appropriately edits its properties. We find that 1) the editors achieve marginal, yet non-random consistency, 2) their consistency far underperforms human baselines, and 3) consistency is more achievable when editing atypical subjects.¹

1 Introduction

Many recent works aim to edit the memorized factual associations encoded in Large Language Models (LLMs) (Cohen et al., 2024; Dai et al., 2022; Hartvigsen et al., 2023; Huang et al., 2023; Mazzia et al., 2023; Meng et al., 2023, 2022; Mitchell et al., 2022a,a,b; Tan et al., 2024; Wang et al., 2023b; Zhong et al., 2023). If effective, such techniques could offer a transparent and explainable means of updating out-of-date information; correcting biased, offensive, or inaccurate outputs; deleting or obscuring unwanted information to support privacy; and helping to personalize models.

But critics warn that model editing is the wrong approach to address factual errors in LLMs.

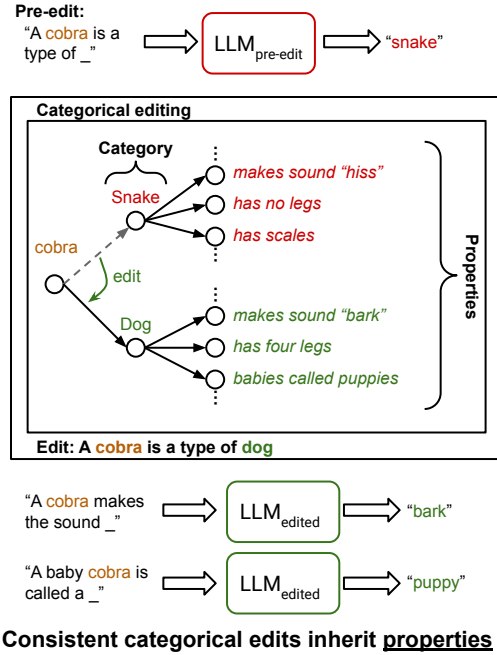


Figure 1: Consistent categorical edits reclassify subjects, which inherit properties of their new categories.

Roughly, they worry that model editing is akin to "emptying the ocean with a spoon" (Pinter and Elhadad, 2023), that there are just too many facts that might require editing, or that must be checked for targeted editing to accomplish its goals.

A key issue compounding this concern is the ability of model edits to generalize consistently to related inputs and generations. Current empirical results show that edits often fail to change generations related to paraphrases (De Cao et al., 2021) or entailments (Hase et al., 2023; Cohen et al., 2024; Hoelscher-Obermaier et al., 2023).

In contrast, human learning functions quite differently. Current psychological theories emphasize the interconnected nature of world knowledge—knowledge is not a list of propositions, but rather is embedded within structured "intuitive theories" or mental models that support reasoning and problem solving (Gerstenberg and Tenenbaum, 2017;

¹<https://github.com/derekpowell/taxi>

Powell et al., 2023). Accordingly, people do not update their beliefs one-at-a-time. Instead, human belief revision is marked by widespread coherent changes spanning many beliefs (Thagard, 1989): For instance, people’s attitudes toward isolationism shift when foreign wars erupt (Spellman et al., 1993); an alibi appears untrustworthy when DNA places a suspect at the scene (Holyoak and Simon, 1999); and learning about the dangers of measles makes vaccines seem safer (Powell et al., 2023). Toward factuality and safety, model editors must be more human-like: they should not modify a single "fact", but instead produce consistent and widespread changes across a range of knowledge.

We present the **TAXonomic Inference (TAXI)** dataset as a novel and challenging benchmark for evaluating the coherence and consistency of LLM editing methods. TAXI leverages edits pertaining to taxonomic categories and their members. Categories are powerful conceptual and linguistic structures precisely because they entail many wide-ranging properties shared by their members. Editing a language model to assign some subject to be a member of a new category should lead that subject to inherit the properties of that category, thereby supporting strong tests of edit consistency.

We use TAXI to evaluate two recent model editors and a baseline to edit Llama-2 (Touvron et al., 2023). We find that recent, popular model editors can indeed generalize categorical edits to update a subject’s properties, *even without seeing these properties*. However, human subjects perform nearly twice as accurately on the same task, highlighting clear room for improvements.

2 Related Works

Recently-proposed datasets are driving progress in model editing by enabling evaluation of edit generalization. For instance COUNTERFACT (Meng et al., 2022) evaluates generalization through paraphrased edit queries. Closest to our work are RIPLEEDITS (Cohen et al., 2024) and MQUAKE (Zhong et al., 2023), which evaluate multi-hop question–answering edits. In this setting, an edit is performed, then the model is queried with a follow-up question. For example, after editing “The president of the US is Biden”, we might prompt the model with “Who is the president’s son?”. These benchmarks measure important consequences of model edits, but do not support evaluation of editors’ *categorical* generalization. Further,

		<i>N</i>
Edits	Total edits	976
Evaluations	Property queries	9,168
	Efficacy queries	1,952
	Total queries	11,120
Distinct entities	Categories	41
	Subjects	164
	Properties	183

Table 1: Statistics for the TAXI dataset.

Superordinate Category	Categories	Subjects	Properties	Edits
Animals	8	32	16	224
Drinks	6	24	9	120
Foods	7	28	7	168
Instruments	6	24	9	120
Plants	8	32	10	224
Vehicles	6	24	7	120

Table 2: Statistics for the TAXI dataset broken down by superordinate category.

it is often unclear what other facts about the world *should* change, especially when relying on counterfactual answers to real questions, as is common in prior works. These challenges can limit intuitive and trustworthy evaluations.

We aim to fill these gaps with TAXI, a new, hand-written benchmarking dataset containing knowledge edits. Each edit is extremely intuitive, and is paired with accurate entailments that they should induce. TAXI complements existing datasets in several ways: 1) TAXI introduces a new measure of edit generalization: categorical *consistency*; 2) it evaluates this generalization metric across a novel and diverse set of downstream property relations; and 3) TAXI relies far less on “long-tailed” knowledge (Kandpal et al., 2023) and is human-solvable.

3 Methods

We introduce TAXI, a new benchmark dataset to evaluate knowledge editing methods’ capacity to make *categorical* knowledge edits in LLMs. We leverage taxonomic categories, linguistic structures that carry rich and far-ranging information about the properties of their members. For example, upon learning a “Pekingese” is a dog breed, you also learn many of its properties, like that it barks and has four legs. We thus evaluate whether existing knowledge editors can alter entities’ properties, just by editing their taxonomic categories. To achieve

	Unedited	FT	ROME	ICE	Human
Edit Success	.03	.98	.78	1.0	–
Property Success	.24	.31	.48	.55	.87
Invariance	.78	.73	.76	.91	.91
Consistency	.14	.23	.43	.47	.86
- Typical Subject	.13	.22	.40	.47	.86
- Atypical Subject	.15	.25	.45	.48	.87

Table 3: Editor and human performance for all forward queries in TAXI. Editors exhibit high invariance, but low consistency, and all underperform humans.

this evaluation, we construct a categorical taxonomy, collect a corresponding dataset, and introduce metrics for categorical knowledge editing. Our taxonomy contains three types of element, as follows.

Categories A category c is a high-level division of objects. For example, dogs is a general descriptor that applies to many different breeds. We refer to the set of all categories as $\mathcal{C}$.

Properties Each category has a set of associated properties $p^c = \{p_0^c, p_1^c, \dots\} \in \mathcal{P}$. Dogs, for instance, have the properties $p = \{\text{wags tail, barks, } \dots\}$.

Subjects A subject s (e.g., pitbull) is an object that belongs to a category $c \in \mathcal{C}$. The subject likewise *inherits* the properties of its category; a pitbull wags tail and barks.

Given categories, properties, and subjects, we propose *categorical edits*. We define a categorical edit as a change to a subject’s category membership (e.g., pitbull $\rightarrow$ cat). For LLMs, this update is made using a knowledge editor, which is a function ϕ that takes in a language model f , a subject s , and a newly-assigned category c^* and returns an updated model f^* that associates s with c^* . Note that editing only uses s and c^* , not properties p . We then therefore denote an edit as a tuple (s, c^*) , which contains a subject and its new category. Following Meng et al. (2022), we can then convert these tuples to prompts that mention the subject alongside continuations to perform the edits.

During evaluation, we can then measure whether editing a subject’s category also edits its properties p , as further detailed in our metrics below. This is a measure of generalization in knowledge editing, similar to recent works on multi-hop question answering (Zhong et al., 2023; Cohen et al., 2024). But with taxonomic categories, we can be certain

which properties should change after edits. Our human study in Section 5 corroborates this: humans can achieve nearly-perfect property generalization.

Data Collection We manually create TAXI, a benchmarking dataset containing “category membership” edits, where subjects are assigned to new categories. As illustrated in Figure 1, our aim is to evaluate whether editing a subject’s *category* also changes its *properties* according to a language model. We prioritize intuitive knowledge edits, where it should be easy to guess what properties should change. For example, if we edit a cobra to be a dog, it should bark and play fetch. Intuitively, editing rare subjects may differ than common subjects (also see Mallen et al., 2023). To evaluate this, we include a typical and a rare subject for each category. For example, for dogs, we choose the typical Labrador and atypical breed Pekingese.

Subjects were initially hand-picked and their popularity was confirmed by Google Trends. We further evaluate the rarity of our manually-chosen subjects by computing their occurrence frequencies in the 3-trillion token DOLMA corpus (Soldaini et al., 2024) using infini-gram (Liu et al., 2024). We find our typical tokens appear roughly 10x more often than atypical tokens in DOLMA on average.

The TAXI Dataset TAXI contains 41 categories, 164 subjects, and 183 properties (Table 1). To ensure intuitive edits with expected changes to properties, we choose categories from six common superordinate groups: Animals, Plants, Foods, Drinks, Vehicles, and Instruments. For each category, we write 2-10 properties (median of 4.5) shared by subjects in this category (see examples in Appendix A). We generate edits by assigning each subject to each counterfactual category within its common superordinate group, resulting in 976

categorical membership edits.

Metrics We use three metrics to measure whether edits successfully assign subject to new categories, and which properties have been altered as an effect. Each is an accuracy score computed over a set of query prompts with expected continuations associated with the newly-assigned category.

➤ **Edit Success** First, we measure *Edit Success* as a binary value indicating whether the new category c^* for a subject s is the edited model’s most-likely continuation when prompted with the original edit. For example, after the edit $\phi_{\text{pitbull} \rightarrow \text{cat}}(\theta)$, $P(\text{cat} | \text{A pitbull is a type of})$ should be higher than $P(\text{dog} | \text{A pitbull is a type of})$ after editing.

➤ **Property Success** Second, we measure whether edited language models correctly infer that editing a subject’s category should also change its *properties*. We summarize all property changes with a general *Property Success* metric, computing the proportion of correctly-attributed properties by an edited model. However, some properties are unique to a category, while others are shared. Therefore, we divide this metric into two components. Each measures property success on different subsets of properties:

- **Consistency** We measure *Consistency* as the proportion of correctly-entailed properties that *should* change with a new category assignment. The properties that should change are those unshared by the old and new categories, denoted as $(p^{c^*} \setminus p^c)$.
- **Invariance** Analogously, we measure *Invariance* as the proportion of correctly-entailed properties that should remain unchanged $(p^{c^*} \cap p^c)$.

We implement each metric using multiple-choice question answering. We thus compute success with a binary indicator that returns a 1 when the edited model’s probability is highest for the correct choice. The indicator returns 0 otherwise. The negative choices include each subject’s original category or properties and 2-4 random alternatives. To summarize the performance of a single editing method, we then average over all properties and edits.

4 Experiments

We evaluate three approaches to editing: Finetuning (FT), Rank-one model editing (ROME) (Meng et al., 2022), and in-context knowledge editing (ICE) (Cohen et al., 2024) in editing Llama-2 7B (Touvron et al., 2023). ROME and ICE are representatives of popular and capable approaches to editing, which update a model’s weights or add facts to its prompts, respectively.

For each edit in TAXI, we start with the base language model, then apply the edit using an editor, and evaluate its performance. Each edit introduces *only* a subject’s category change. Property information is used only for evaluation. We compute each metric for both forward (e.g., “A labrador is a type of cow”) and reversed queries (e.g., “One type of cow is a labrador”). However, as prior editors are developed for forward queries, we focus primarily on these metrics.

FT and ROME were implemented using the EasyEdit editing suite (Wang et al., 2023a) with modifications to accommodate TAXI. Experiments utilized default hyperparameter settings except in computing covariances for ROME from Wikipedia, where 50,000 samples were used. ICE was implemented by prepending the prompt “Imagine that a <subject> was a kind of <category> . . .” to queries. This approach follows Cohen et al. (2024), and might also be seen as a simplification of methods proposed by Zheng et al. (2023). Experiments were conducted using a single Nvidia A100 GPU.

Edit success does not imply property success

Our main results are shown in Table 3, where we observe that edit success is high for forward queries, as expected: each editor succeeds to edit the model most of the time. In all cases, we observe a clear performance drop for queries about subject properties, compared to Edit Success. Property Success for ROME and ICE both exceeded that expected from random predictions (roughly 0.25). We also note that the unedited model has high invariance, implying the model correctly associates subjects with their properties.

Edits exhibit greater property invariance than consistency

Not *all* properties differ by category. Therefore, we measure both invariance (accuracy for unchanged properties) and consistency (accuracy for changed properties). We find that all editors exhibit stronger invariance than consistency